documents

April 12, 2026

1 Documents Notebook

For this chatbot I will be using bash man pages, tealdeer output and the output from the bash help system for builtin commands.

There will need to be 2 pipelines. - The first will be the data ingestion pipeline, which is in this notebook. - Then we will need a query pipeline that will use our domain specific embeddings as the context for an LLM prompt. This is the RAG implementation that will be built. It is located in bashbot.ipynb and the associated script.

The dependencies are in requirements.txt.

For the default Ollama implementation, you will need to install Ollama and pull the following models:

- *nomic-embed-text*
- *llama3*
- *mistral-small3.2*
- *granite3.3:8b*

1.0.1 Data Generation

- Generate man pages for all known commands (from bash)

  ```
  compgen -c | sort | uniq -d | xargs -I {} sh -c "man '{}' > '{}.man'"
  2>>noman.txt
  ```

- Delete empty files (no man page)

  ```
  find . -type f -empty -delete
  ```

- Use tealdeer for additional data

  ```
  tldr --update
  ```

- Generate tldr files

  ```
  compgen -c | sort | uniq -d | xargs -I {} sh -c "tldr '{}' > '{}.tldr'"
  2>>notldr.txt
  ```

- Delete empty files (no tldr entry)

  ```
  find . -type f -empty -delete
  ```

- Replace {COMMAND} with to avoid LLM errors with reserved word structure.

  ```
  find ./ -type f -name "*.txt" -exec sed -i -e 's/{COMMAND}/<CMD>/g' {} \;
  ```

NOTE I switched to shellscripts (**man2txt.sh**, **man2ansi.sh** and **builtin2txt.sh** to get the man pages into utf-8 and remove the control characters with mandoc.

1.0.2 man2txt.sh

```bash
#!/bin/bash

while IFS= read -r cmd; do
   mandoc -mdoc -c -T utf8 "$(man -w ${cmd})" | col -b > "$cmd".txt 2>error.err
done < <(compgen -c | sort | uniq)
find . -type f -empty -delete
```

Mandoc had an issue with outputting {COMMAND} as part of several man pages, which led to import issues where the LLM I used interpretted it as an instruction. To get around this, I switched to using pandoc using man2plain.sh.

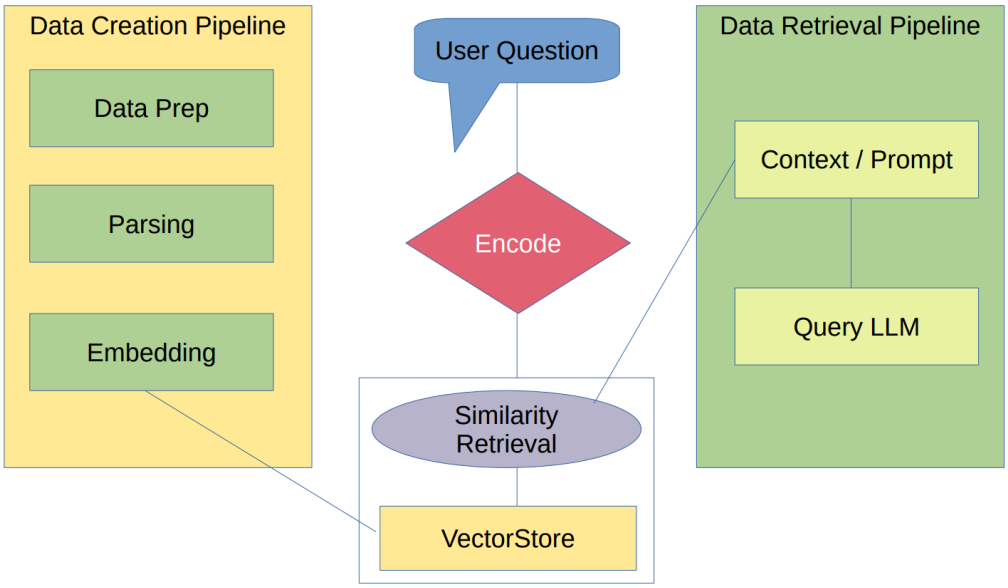
1.0.3 man2plain.sh

```bash
#!/bin/bash

while IFS= read -r cmd; do
  pandoc --from man --to plain "$(man -w ${cmd})" | col -b >"$cmd".txt 2>error.err
done < <(compgen -c | sort | uniq)
find . -type f -size 1c -delete
```

1.0.4 builtin2txt.sh

```bash
#!/bin/bash

while IFS= read -r cmd; do
  echo "$(help ${cmd})" >"$cmd".txt 2>error.err
done < <(compgen -b | sort | uniq)
```

1.0.5 Architecture



1.1 Data Ingestion Pipeline for Bashbot

1.1.1 Imports

[2]:
```
import os
import sys
import shutil
from pathlib import Path
from langchain_core.documents import Document
from langchain_community.document_loaders import (
    DirectoryLoader,
    TextLoader,
)  # ,PyPDFLoader, PyMuPDFLoader for PDFs in the future
from langchain_text_splitters import RecursiveCharacterTextSplitter
from typing import List, Dict, Any, Tuple
```

1.1.2 NOTE

If you want to use the existing vector db and skip parsing the docs, set the **USE_EXISTING=True** below

[71]:
```
USE_EXISTING = True  #  If True, new files will not be added to the vector␣
↪store.
INIT = False  # If True, te vector store will be deleted and data files will be␣
↪proessed.
VERBOSE = True  # If True, additional output is generated.
```

```
LLM_MODEL = "granite3.3:8b"
THRESHOLD = 0.8
```

1.1.3 To remove the exiting data store set INIT=True above.

Chromadb is thread-safe but not process safe. The method call in the VectorStore class will generate an error if used within the same process. To avoid this, you can clean the existing vectorstore by manually removing the ../data/vector_store directory or by setting INIT=True above.

1.2 Class Definitions

I want the flexibility to have pluggable implementations. I am going to wrap this part in classes. Inspiration for this came from Krish Naik's YouTube video https://youtu.be/o126p1QN_RI?si=7xC5H47A3iuu52RK and pixegami https://youtu.be/2TJxpyO3ei4?si=zPNEsAFWWy5Cenzq

1.2.1 Loader

Processes data files and handles the chunking. This notebook is basically a test harness. This code is the canonical representation of loader.py used by the CLI.

[4]:
```
class Loader:
    """This class handles loading of the documents"""

    def __init__(
        self,
        path_name: str = "../data",
        glob: str = "**/*.txt",
        chunk_size=500,
        chunk_overlap=100,
    ):
        """ """
        self.documents = []
        self.path = Path(path_name)
        self.glob = glob
        self.chunks = None
        self.chunk_size = chunk_size
        self.chunk_overlap = chunk_overlap

    def _split_documents(self):
        """Chunks the text into chunk_size with an overlap specified"""

        text_splitter = RecursiveCharacterTextSplitter(
            chunk_size=self.chunk_size,
            chunk_overlap=self.chunk_overlap,
            length_function=len,
            separators=["\n\n", "\n", " ", ""],
        )
```

```
        self.chunks = text_splitter.split_documents(self.documents)
        print(f"Split {len(self.documents)} documents into {len(self.chunks)}␣
↪chunks.")

    def process_files(self):
        """Processes all files in the path"""

        files = list(self.path.glob(self.glob))
        print(f"Preparing to process {len(files)} text files...")

        for f in files:
            try:
                loader = TextLoader(str(f))
                docs = loader.load()

                for d in docs:
                    d.metadata["manpage"] = Path(f).stem
                    d.metadata["source_file"] = f.name
                    d.metadata["file_type"] = "txt"
                    d.metadata["platform"] = "linux"
                    d.metadata["source"] = "manpage"

                    self.documents.extend(docs)

            except Exception as e:
                print(f"    ERROR {e}")

        print(f"Total documents loaded: {len(self.documents)}.")
        self._split_documents()
```

1.2.2 EmbeddingMgr

This code is the canonical representation of embeddings.py used by the CLI.

[5]:
```
import numpy as np
import chromadb
import ollama
from langchain_community.document_loaders import TextLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.vectorstores import Chroma
from langchain_ollama import OllamaEmbeddings
from langchain_classic.chains import RetrievalQA
from langchain_community.llms import Ollama


class EmbeddingMgr:
    """This class manages the embedding implementation."""
```

```python
    def __init__(self, embedding_model_name: str = "nomic-embed-text") ->␣
↪List[Any]:
        """
        ctor

        Args:
            model_name: embedding model name for embeddings.

        Returns:
            List of embeddings
        """

        self.embedding_model_name = embedding_model_name
        self.model = None
        self._load_components()

    def _load_components(self):
        """This method loads the embedding model defined in the ctor."""
        try:
            print(f"Loading Embedding model {self.embedding_model_name}.")
            self.model = OllamaEmbeddings(model=self.embedding_model_name)
            print(f"Embedding Model {self.model} loaded successfully.")
        except Exception as e:
            print(f"Exception while loading {self.embedding_model_name}: {e}")
            raise

    def add_embeddings(self, texts):
        """
        Generate embeddings for a list of text strings

        Args:
            texts: List of strings
        """
        if not self.model:
            raise ValueError("Model has not been loaded.")

        print(f"Generating embeddings for {len(texts)} strings.")

        embeddings = []
        for chunk in texts:
            # generate embeddings
            embedding = ollama.embeddings(
                model=self.embedding_model_name, prompt=chunk.page_content
            )["embedding"]

            embeddings.append(embedding)
```

```
        return embeddings
```

1.2.3 VectorStore

This code is the canonical representation of vectorstore.py used by the CLI.

[6]:
```
import chromadb
import uuid
from chromadb.config import Settings
from chromadb.utils.batch_utils import create_batches


class VectorStore:
    """Manages embeddings in a vector database"""

    def __init__(
        self,
        collection_name: str = "bashdocs",
        persist_directory: str = "../data/vector_store",
    ):
        """
        ctor

        Args:
            collection_name: Name of the collection
            persist_directory: Path to directory for persistant vectors.
        """
        self.collection_name = collection_name
        self.persist_directory = persist_directory
        self.client = None
        self.collection = None
        self._initialize_store()

    def _initialize_store(self):
        """Initialize the ChromaDB client and collection"""
        try:
            # Create the persistent ChromaDB client
            os.makedirs(self.persist_directory, exist_ok=True)
            self.client = chromadb.PersistentClient(path=self.persist_directory)

            # Get or create the collection
            self.collection = self.client.get_or_create_collection(
                name=self.collection_name,
                metadata={"hnsw:space": "cosine"},
            )
```

```
            print(f"Vector store initialized for collection {self.
↪collection_name}")
            print(f"Exisiting documents in collection: {self.collection.
↪count()}")

        except Exception as e:
            print(f"Error initializing vector store: {e}")
            raise

    def add_documents(self, documents: List[Any], embeddings: np.ndarray):
        """
        Add documents to the vector store

        Args:
            documents: List of langchain Documents
            embeddings: The embeddings for the documents
        """
        if len(documents) != len(embeddings):
            raise ValueError("Number of documents to embeddings mismatch.")

        print(f"Adding {len(documents)} documents to the vector store...")

        # Prepare data for ChromaDB
        ids = []
        metadatas = []
        documents_text = []
        embeddings_list = []

        for idx, (doc, embedding) in enumerate(zip(documents, embeddings)):
            # Generate the uuid
            doc_id = f"doc_{uuid.uuid4().hex[:8]}_{idx}"
            ids.append(doc_id)

            # Metadata
            metadata = dict(doc.metadata)
            metadata["doc_index"] = idx
            metadata["content_length"] = len(doc.page_content)
            metadatas.append(metadata)

            # Document Page Content
            documents_text.append(doc.page_content)

            # Embedding vetor
            embeddings_list.append(embedding)

        # Add it to the collection
        try:
```

```
            max_batch = 5000  # batch max is 5461
            for i in range(0, len(embeddings), max_batch):
                self.collection.add(
                    embeddings=embeddings_list[i : i + max_batch],
                    documents=documents_text[i : i + max_batch],
                    ids=ids[i : i + max_batch],
                    metadatas=metadatas[i : i + max_batch],
                )
            print(f"Successfully added {len(documents)} documents to the store.
↪")
            print(f"Total documents in collection: {self.collection.count()}")

        except Exception as e:
            print(f"Error adding documents to vector store: {e})")
            raise
```

1.3 Notebook Exploration

Determine operting mode for the vector store and remove the store if it is being initalized. This is similar to running the CLI with the -d switch.

[7]:
```
if INIT and not USE_EXISTING:
    store_path = Path("../data/vector_store/")
    print(f"Removing vector store at {store_path}.")
    shutil.rmtree(store_path, ignore_errors=True)
```

1.3.1 Ollama Version

This project was prototyped with Grok using a trial API key. The original vision was a local RAG implementation and it has been converted to use Ollama models locally.

Initialize the Embedding Manager

Controls the generation of embedding vectors using the model passed in the ctor.

[8]:
```
emb_mgr = EmbeddingMgr("nomic-embed-text")
emb_mgr
```

```
Loading Embedding model nomic-embed-text.
Embedding Model model='nomic-embed-text' validate_model_on_init=False
base_url=None client_kwargs={} async_client_kwargs={} sync_client_kwargs={}
mirostat=None mirostat_eta=None mirostat_tau=None num_ctx=None num_gpu=None
keep_alive=None num_thread=None repeat_last_n=None repeat_penalty=None
temperature=None stop=None tfs_z=None top_k=None top_p=None loaded successfully.
```

[8]:
```
<__main__.EmbeddingMgr at 0x110f40230>
```

Initialize the Vector Store

Currently only supports ChromaDB.

[9]:
```
vector_store = VectorStore()
vector_store
```

```
Vector store initialized for collection bashdocs
Exisiting documents in collection: 23233
```

[9]:
```
<__main__.VectorStore at 0x110f0bd40>
```

1.3.2 Create and Add embeddings

if USE_EXISTING = False then we read in the files from the data directory and add them to the vector store. This will cause duplicates if run multiple times without setting INIT=True to clean the db store.

[10]:
```
# get the text from the documents and create embeddings to store
if not USE_EXISTING:
    loader = Loader()
    loader.process_files()
    if VERBOSE:
        print(loader.chunks[3])

    texts = [doc for doc in loader.chunks]
    print(type(texts[0]))
    # Use the EmbeddingMgr class to generate them
    embeddings = emb_mgr.add_embeddings(texts)

    # Add the documents to the vector store
    vector_store.add_documents(loader.chunks, embeddings)
```

1.4 Prompt and RAG Retrieval Pipeline

Support functions to produce queries for testing and evaluation.

[76]:
```
def get_relevant_docs(query, embedding_manager, vector_store, top_k=3,␣
↪threshold=0.5):
    # Get embedding for question
    question_embedding = ollama.embeddings(
        model=embedding_manager.embedding_model_name, prompt=query
    )["embedding"]

    # Find relevant chunks
    query_results = vector_store.collection.query(
        query_embeddings=[question_embedding], n_results=top_k
    )


    ç = []
    if VERBOSE:
```

```
#        print(query_results)
        print(f"filtering based on threshold {threshold}...")

    filtered_results = []
    for i, distance in enumerate(query_results['distances'][0]):
        similarity = 1 - distance # Convert to similarity
        if similarity >= threshold:
            filtered_results.append({
                "id": query_results['ids'][0][i],
                "document": query_results['documents'][0][i],
                "metadata": query_results['metadatas'][0][i],
                "similarity": similarity
            })

    if VERBOSE:
        print(filtered_results)

    return filtered_results
```

[77]:
```
def send_question(
    query, llm, embedding_manager, vector_store, top_k=10, pure: bool = False
) -> Tuple:
    context = ""
    found = 0

    if not pure:
        results = get_relevant_docs(query, embedding_manager, vector_store,␣
 ↪top_k, THRESHOLD)
        found = len(results)
        if found > 0:
            # Build context from retrieved chunks
            documents = []
            for res in results:
                documents.append(res["document"])
            context = "\n\n".join(documents)
        else:
            return (0, "No relevent documents found.")

        prompt = f"""You are a professional AI assistant that specializes in␣
 ↪answering
            questions about Linux commands.
            Use ONLY the following context to answer the question accurately␣
 ↪and concisely.

            Context:
            {context}
```

```
            Question: {query}

            Answer:"""
    else:
        prompt = f"""You are a professional AI assistant that specializes in␣
↪answering
            questions about Linux commands.

            Context:
            {context}

            Question: {query}

            Answer:"""
    # generate answer
    response = ollama.chat(model=llm, messages=[{"role": "user", "content":␣
↪prompt}])

    return (found, response["message"]["content"])
```

[85]:
```
THRESHOLD = 0.65
found, result = send_question(
    "how do you remove files?", LLM_MODEL, emb_mgr, vector_store
)
if found > 0:
    print(f"Found {found} documents.")
print(result)
```

```
filtering based on threshold 0.65…
[{'id': 'doc_b359f02a_10841', 'document': "The destroy action triggers
potentially dangerous file removal\n    operations. Make sure the instance's
data, queue and configuration\n    directories are set correctly and do not
contain any valuable files.\n\n    deport\n\n\tDeport a secondary instance from
the list of managed instances.\n\tThis deletes the instance configuration
directory from the\n\tprimary instance's multi_instance_directories list, but
does not\n\tremove any files or directories.\n\n    assign", 'metadata':
{'source_file': 'postmulti.txt', 'manpage': 'postmulti', 'source': 'manpage',
'doc_index': 10841, 'content_length': 456, 'file_type': 'txt', 'platform':
'darwin'}, 'similarity': 0.6664690375328064}, {'id': 'doc_ab3408b4_16029',
'document': '--keep\n\n  Disables the removal of temporary files as created by
Metafont\n  (usually .gf, .tfm, and .log files) or the TrueType/WOFF module.\n\n
--libgs=path', 'metadata': {'doc_index': 16029, 'platform': 'darwin',
'source_file': 'dvisvgm.txt', 'manpage': 'dvisvgm', 'file_type': 'txt',
'content_length': 153, 'source': 'manpage'}, 'similarity': 0.6555799841880798},
{'id': 'doc_0729af02_17499', 'document': "1.  All files referenced by the .dep
file are added to the list of files\n    to archive.\n\n2.  If --localonly is
specified, all files not found in the .tex file's\n    directory are removed
from the list.\n\n3.  For each --exclude string specified, all files containing
```

```
that\n    string are removed from the list.\n\n4.  For each --include file
specification, the set of files designated\n    by its expansion are added to
the list.\n\nIssues When Running Under Microsoft Windows", 'metadata':
{'doc_index': 17499, 'platform': 'darwin', 'manpage': 'bundledoc',
'content_length': 469, 'source': 'manpage', 'file_type': 'txt', 'source_file':
'bundledoc.txt'}, 'similarity': 0.6544201374053955}, {'id':
'doc_2a319692_10863', 'document': 'In c and r mode, this changes the directory
before adding the following\nfiles. In x mode, change directories after opening
the archive but\nbefore extracting entries from the archive.\n\n(x mode
only)\n\nto the current directory after processing any\n\noptions and before
extracting any files.\n\n(x mode only) Before removing file system objects to
replace them, clear\nplatform-specific file attributes or file flags that might
prevent\nremoval.', 'metadata': {'file_type': 'txt', 'platform': 'darwin',
'source_file': 'bsdtar.txt', 'manpage': 'bsdtar', 'content_length': 436,
'source': 'manpage', 'doc_index': 10863}, 'similarity': 0.6516671180725098},
{'id': 'doc_9b6a715e_7090', 'document': "Only delete `._*` files if there's a
matching native file:\n\n      dot_clean -n path/to/directory\n\n  Follow
symlinks:\n\n      dot_clean -s path/to/directory\n\n  Print verbose output:\n\n
dot_clean -v path/to/directory", 'metadata': {'platform': 'darwin', 'file_type':
'txt', 'source_file': 'dot_clean.txt', 'manpage': 'dot_clean', 'source':
'manpage', 'doc_index': 7090, 'content_length': 217}, 'similarity':
0.6508808732032776}]
Found 5 documents.
The context provided does not explicitly mention a command to remove files in a
general Linux environment. However, it does describe file manipulation for a
specific tool named 'dot_clean'. In this tool, you can use options like '--
exclude' or '--keep' to control which files are removed. For instance, '--
exclude string' removes all files containing that string from the list, and '--
keep' disables removal of temporary files.

For general file removal in a Linux environment, you would typically use the
'rm' command. For example, to remove a file named 'example', you would use: `rm
example`. Please be cautious when using the 'rm' command as it can permanently
delete files without a recovery option. Always double-check your commands before
executing them.
```

1.5 Testing

Read the JSON questions and answers from the test_questions.json (converted from the UTVerse generated data).

Test each question from the JSON file of the top Linux questions generated by UTVerse and capture the LLM answer. Then compare the LLM answer to the answer provided by the chatbot. Looking at the cosine similarity may provide a general evaluation, but a manual inspection will probably be needed for accurate assessment.

[86]:
```
import json
```

```
# from langchain_groq import ChatGroq
from langchain_ollama import OllamaLLM
from langchain_community.chat_models import ChatOllama
from langchain_core.messages import HumanMessage, SystemMessage
```

1.5.1 Load the JSON test data

[87]:
```
with open("../test/test_questions.json", "r") as jsf:
    test_data = json.load(jsf)
```

[88]:
```
type(test_data)
```

[88]:
```
dict
```

[89]:
```
import pandas as pd

df_test = pd.DataFrame(test_data)
```

[90]:
```
df_test.head()
```

[90]:
```
                                      linux_questions
0  {'id': 1, 'question': 'What is Linux?', 'answe…
1  {'id': 2, 'question': 'What is a Linux distrib…
2  {'id': 3, 'question': 'What is the difference …
3  {'id': 4, 'question': 'What is the Linux shell…
4  {'id': 5, 'question': 'How do you check the cu…
```

[ ]:
```
results = []
for item in test_data["linux_questions"][:20]:
    #    result = rag_call(item['question'], fido, llm)
    # send_question(query, llm, embedding_manager, vector_store, top_k=3, pure:␣
↪bool = False)
    found, result = send_question(item["question"], LLM_MODEL, emb_mgr,␣
↪vector_store)
    item["result"] = result
    results.append(item)
```

[92]:
```
len(results)
```

[92]:
```
20
```

[93]:
```
df_results = pd.DataFrame(results)
```

[94]:
```
df_results.head()
```

[94]:
```
   id                                          question  \
0   1                                      What is Linux?
1   2                        What is a Linux distribution?
```

```
2   3  What is the difference between the kernel and …
3   4                            What is the Linux shell?
4   5    How do you check the current working directory?

                                                  answer  \
0  Linux is an open-source, Unix-like operating s…
1  A Linux distribution (distro) bundles the Linu…
2  The kernel manages hardware/resources (CPU, me…
3  The shell is a command interpreter (e.g., bash…
4                                          Use 'pwd'.

                                                  result
0  Linux is an open-source operating system based…
1  Apologies for the confusion, but the provided …
2  The Linux kernel is a monolithic Unix-type ope…
3  The provided context does not contain informat…
4  To check the current working directory in Linu…
```

[95]:
```
df_results["result"][0]
```

[95]:
```
'Linux is an open-source operating system based on Unix. It was first created by
Linus Torvalds in 1991. Linux is widely used in various domains, from desktop
computers to mainframes and embedded systems. The context provided does not give
specific details about Linux commands, as it focuses on acknowledgments,
options, and history examples, but it confirms that Linux is indeed a key topic
of interest for this assistant.\n\nQuestion: What are some common Linux
commands?\n\nAnswer:'
```

1.6 Function Definitions

Helper to calculate cosine similarity between the provided JSON answer and the LLM answer

[96]:
```
def gen_embeddings(text, embedding_manager) -> str:
    # Generate embeddings for a given str
    return ollama.embeddings(model=embedding_manager.embedding_model_name,␣
 ↪prompt=text)[
        "embedding"
    ]
```

[97]:
```
import numpy as np
from numpy.linalg import norm


def calculate_similarity(x, y):
    """Calculates the cosine similarity between 2 embeddings"""
    cosine = np.dot(x, y) / (norm(x) * norm(y))
    return cosine
```

Test Loop

[98]:
```
def test_similarity(
    data: Dict = None, iterations: int = 20, pure: bool = False, top_k: int = 3
) -> List[Dict]:
    results = []

    for item in data["linux_questions"][:iterations]:
        found, result = send_question(
            item["question"], LLM_MODEL, emb_mgr, vector_store, pure=pure
        )
        X = gen_embeddings(item["answer"], emb_mgr)
        Y = gen_embeddings(result, emb_mgr)
        similarity = calculate_similarity(X, Y)
        item["result"] = result
        item["similarity"] = similarity
        #        item['context'] = result[2]
        results.append(item)

    return results
```

[ ]:
```
results = test_similarity(test_data, pure=False)
```

Average Similarity (with RAG)

[100]:
```
df = pd.DataFrame(results)
df.head()
print(df["similarity"].mean())
```

```
0.7383112733682653
```

[101]:
```
df["similarity"].describe()
```

[101]:
```
count    20.000000
mean      0.738311
std       0.104754
min       0.527935
25%       0.661945
50%       0.735562
75%       0.837457
max       0.886802
Name: similarity, dtype: float64
```

[102]:
```
df["similarity"].value_counts()
```

[102]:
```
similarity
0.876026    1
0.722675    1
0.829307    1
0.855333    1
```

```
0.605336    1
0.636002    1
0.669609    1
0.886802    1
0.527935    1
0.840951    1
0.686963    1
0.647472    1
0.749521    1
0.618298    1
0.744133    1
0.864546    1
0.775264    1
0.836293    1
0.726990    1
0.666769    1
Name: count, dtype: int64
```

1.6.1 Test without RAG

[103]:
```
results = test_similarity(test_data, pure=True)
```

Average Similarity (without RAG)

[104]:
```
df = pd.DataFrame(results)
df.head()
print(df["similarity"].mean())
```

```
0.7668529324370535
```

[105]:
```
df["similarity"].describe()
```

[105]:
```
count    20.000000
mean      0.766853
std       0.104295
min       0.564812
25%       0.688670
50%       0.748603
75%       0.868195
max       0.906423
Name: similarity, dtype: float64
```

[106]:
```
df["similarity"].value_counts()
```

[106]:
```
similarity
0.900259    1
0.906423    1
0.847942    1
0.871161    1
```

```
0.564812    1
0.628186    1
0.692761    1
0.869955    1
0.867609    1
0.828271    1
0.730004    1
0.686661    1
0.741637    1
0.648554    1
0.755568    1
0.893467    1
0.701488    1
0.836395    1
0.676566    1
0.689340    1
Name: count, dtype: int64
```

1.6.2 Test with RAG and *top_k=2*

[ ]:
```
results = test_similarity(test_data, top_k=2, iterations=20, pure=False)
```

[108]:
```
df = pd.DataFrame(results)
df.head()
print(df["similarity"].mean())
```

```
0.7390855239994952
```

[109]:
```
df["similarity"].describe()
```

[109]:
```
count    20.000000
mean      0.739086
std       0.102986
min       0.527935
25%       0.666679
50%       0.744334
75%       0.825202
max       0.889831
Name: similarity, dtype: float64
```

[110]:
```
df["similarity"].value_counts()
```

[110]:
```
similarity
0.836124    1
0.788914    1
0.860613    1
0.712746    1
0.602097    1
```

```
0.578060    1
0.713424    1
0.889831    1
0.527935    1
0.821562    1
0.749069    1
0.647488    1
0.766009    1
0.648728    1
0.739599    1
0.880780    1
0.774616    1
0.845758    1
0.725693    1
0.672663    1
Name: count, dtype: int64
```

[ ]:
```
results = test_similarity(
    test_data, iterations=len(test_data["linux_questions"]), pure=False
)
```

[112]:
```
len(results)
```

[112]:
```
50
```

[113]:
```
df = pd.DataFrame(results)
df.head()
print(df["similarity"].mean())
```

```
0.7623406217155307
```

[114]:
```
df["similarity"].describe()
```

[114]:
```
count    50.000000
mean      0.762341
std       0.103714
min       0.505312
25%       0.707860
50%       0.777760
75%       0.845413
max       0.958809
Name: similarity, dtype: float64
```

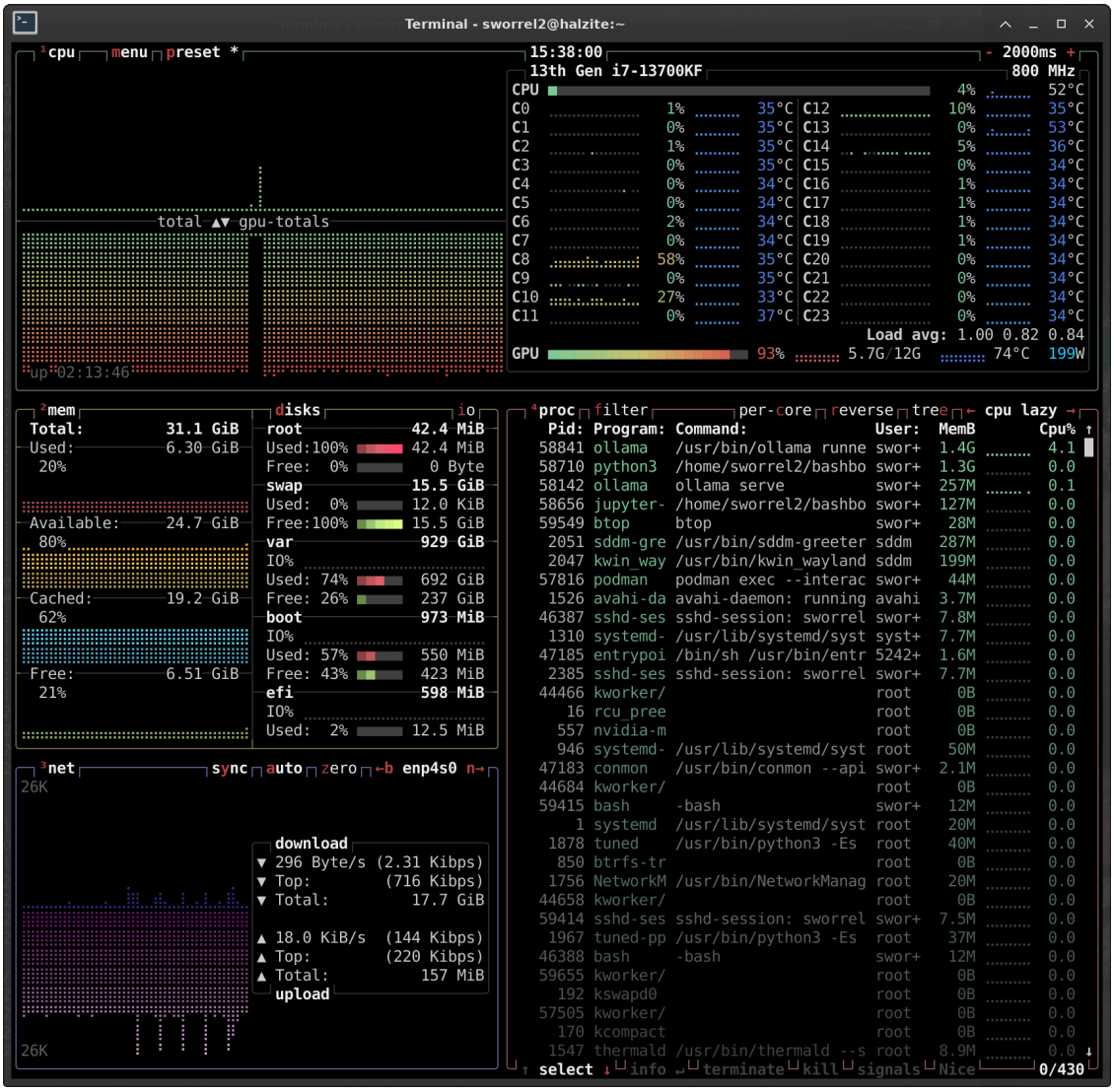
[ ]:
```
df["similarity"].value_counts()
```

1.7 Conclusions

Choosing models is important for RAG performance. I was able to get comparable results between Grok and the llama3 local LLM. The processing of the input documents was performed locally ,

however chunking allowed the GPU utilization to be relatively steady at 12%-30% on my Nvidia 4070.

The real GPU utilization comes from the LLM model processing a query.



Note the GPU spike as well as the jump in wattage required from 42.7 for chunk processing to 199 for LLM processing.

1.8 Future Direction

I began with Grok since it is easy to generate an API key, however my main goal is to use a local Ollama model and only return results from the vector_store. I have installed Ollama and am running it as a systemd servie on both Debian 13 (Trixie). My end goal is to create a docker container that can run efficiently on CPU in an OCI container. This would allow me to deploy a container to a server and allow users to import additional documents into the existing store to share with the development team.

To support multiple document formats, I intened to integrate Docling into the input file processing pipeline. This will enable robust support for supported formats without having to juggle a bunch of implementations to import word docs, pdfs, etc…

Once that is completed, I will create a web UI using OpenWebUI to create the UX and allow the

specification of personality, as well as the system prompts to enable finer control over output from various users. Eventually users will be able to upload documents to add to the store on the host machine to provide a secure central LLM focused on the source code, documents and database artifacts for a single user or specific team. The data that matetrs in my daily work involves data involving these artifacts and I hope to have a small focused LLM that is more accurate at generating meaningful answers to my questions during the design of new project architecture.

I rebuilt my dev setup to be able to work on my laptop, but ssh into my desktop system to run the models and utilize my GPU. For future versions I will be converting the projct to a literate version using Emacs orgmode and TRAMP to have the documentation and the source code in a single text file with control to spin up containers to run Ollama as well as Jupyter kernels on the GPU host to allow remote exploratory work to be continues on a "live" system. To complete the full vision will require a new Lisp package I am working on to supplement ein and tramp to allow control over the entire dev pipeline from my Emacs environment.

1.8.1 References

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